Impact of Active Aero on Tire loads, Fuel consumption and Speed of a 2014 McLaren P1

Graduate Research Project

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1 Project Proposal

EFFECT OF ACTIVE AERO COMPONENTS ON VEHICLE PERFORMANCE AND TIRE PARAMETERS

Active Aero refers to the use of movable components on the body of a vehicle to "actively" manipulate the flow of air around it. At high speeds, vehicles experience considerable resistance due to opposing air flow (drag), which can be significantly reduced through the use of these movable components. The benefits of active aero are not, however, limited only to performance in a straight line. They also extend to improved braking, better cornering and higher fuel efficiency, all factors which can greatly improve a car's lap time.

The effects of these changes on a car's overall performance are what I intend to explore in this study, such as changes in drag coefficients, efficiency, braking distance, normal load acting on the car (downforce). I also intend to research the effects of active aero on various tire parameters, such as changes in lateral and longitudinal forces on the tire, loaded radius, slip ratio, resulting grip on road surface.

To this end, I will be looking into the various different components that fall under the umbrella of active aero and see what each of them contribute to a vehicle's performance. This will consist of researching their design and what parameters they directly affect. I will also look at existing cases of active aerodynamics in motorsport to see the immediate performance gains. My study will be based around the component which has the most noticeable impact on car performance.

Using the results from this analysis, I will determine the new forces acting upon the tire of a car and compare them to the same tire on the same car without active aero using MATLAB simulations. This data can be used in the simulation of a race consisting of straights and corners to test the full range of the additional component.

Through this project, I hope to ascertain the degree to which active aero improves car performance and the resulting effect on tire performance. While we are aware that active aero components improve vehicle performance on many dimensions, I believe seeing the quantitative effects of just a single component will give a better idea of the scope of this technology. Along the way, I also hope to find applications of active aero in daily use cars instead of just for on track performance.

2 Literature Review

PAPER 1: Review of Research on Vehicles Aerodynamic Drag Reduction Method
This paper's focus was centered around aerodynamic control as a means of mitigating fuel consumption in various ground vehicles. The objective was to reduce drag forces caused by separation of air flow over and behind the vehicle, since a considerable portion of energy from the engine is used in overcoming aerodynamic drag.
The paper goes into great detail about various methods of active aerodynamic systems, the percentage of performance improvement in terms of reduced drag coefficient. It also explored types of passive aerodynamic control and measured the results against the improvement that was found in the active counterparts. Since I intend to focus on active aero for this research project, I will not go into too much detail about the passive flow control systems. The paper reviews different methods of aerodynamic flow control and outlines the reduction in drag coefficient of the vehicle from before and after the application of the system.

The components are classified into:
Active Flow Control: These systems rely on an external source of energy and a control system to modulate the flow of air around the vehicle. These components can be adjusted by the pilot into positions so they can optimally guide the flow of air around the vehicle under different circumstances. An example of this is movable flaps on rear wings of Formula race cars.
Passive Flow Control: These systems manipulate the flow of air around the vehicle without any need for a power source, These are stationary so once installed, they cannot change the drag coefficient. An example of this would be a front diffuser on sports cars.

The analyses of these methods were performed over varying vehicle speeds and geometries. The active flow control techniques studied were:

- Pulse Jets: Mechanisms placed on the rear of a vehicle that periodically inject pressurized air into the regions of separated flow, reducing the resultant drag forces.
- Steady Blowing: Induces a stream of high-energy air into the boundary layer surrounding a vehicle and thus, adds momentum to the slower-moving fluid near the surface. This helps the fluid resist pressure gradients which results in separation.
- Suction: Removes low-energy fluid near the surface of the vehicle, reducing the boundary layer's thickness. The removal of this slower air reduces the effects of pressure differences that would lead to flow separation.

PAPER 2: A Review of Active Aerodynamic Systems for Road Vehicles

This paper focuses on the effect of active aerodynamics in vehicle track performance as well as comfort for daily commuters. For performance, this study inspects active aerodynamic systems and how they adapt to changing conditions such as speed, yaw rate, and braking. This review also explores historical, current, and potential cases of active aerodynamics and how one might adapt similar/improved systems onto land vehicles.

The paper explored some detailed mathematical simulations that find the degree to which active aero components affected things like ride comfort and vehicle handling. For this, the authors examined interactions between aerodynamic forces and vehicle motion using Fluid Structure Interactions (FSI). By considering vehicle body pitch, roll, and suspension deflection in their simulations, these models showed how changes in aerodynamic loads affect tire forces and, as a result, overall stability. For example, the coupling of downforce from adjusting a flap angle with suspension deflection was used to understand performance during cornering and braking.

Among the many analysis models used in the paper were nonlinear models, using Pacejka's Magic Formula. The magic formula is an empirical relation that can simulate tire behavior under combined slip and lateral load conditions. This model factors in parameters like camber and slip angles to analyze active aero effects on cornering stiffness and lateral force generation.

The last method discussed in the paper is Coupled Simulation models. They work by combining CFD with mechanical models and help with analysis of changing aerodynamic effects. Simulation using overset grid techniques (overlapping meshes) and deformable meshes allow for tracking the moving surfaces attacked by air flow. All these models mentioned and their results were shown to be validated through wind tunnel tests and track data.

PAPER 3: Numerical Investigation of the Influence of Tire Deformation and Vehicle Ride Height on the Aerodynamics of Passenger Cars

This paper focuses on how tire properties like deformation and vehicle ride height contribute to aerodynamic drag.

The paper looked into two vehicle models: DrivAer and AeroSUV, two analysis models frequently used in wind tunnel testing. These models were analyzed with various ride heights and tire deformation levels. The investigation was carried out using unsteady Computational Fluid Dynamics (CFD) simulations, applying vertical movements to adjust the necessary parameters.

The study simulated three tire deformation levels: baseline (no deformation), medium, and high deformation. Higher deformation levels caused a reduction in drag for the front tires but an increase for the rear ones, increasing overall vehicle drag. This result was to be expected as one can intuit an increase in drag around a corner.

The study also varied vehicle ride height in increments of +30 mm (lifted) and-30 mm (lowered) and observed that lowering the ride height reduced aerodynamic drag, whereas increasing it led to an increase in drag. The effect was

more pronounced for the detaile underbody configuration.
When changes in tire deformation and ride height were executed together, tire deformation dominated the increase in drag, while the interaction between the ride height and the tire deformation showed that lowering the ride height could decrease some of the drag caused by tire deformation. Overall, deformation remained the stronger contributor. The findings were validated by comparing CFD results with experimental data for the DrivAer and AeroSUV.

CONCLUDING REMARKS

The papers I selected for this review aligned with the goals and desired learnings of this project. Two of them were chosen to learn about active aerodynamic systems, their impact on various vehicle parameters like overall speed, fuel consumption, cornering stability, and ride comfort. The last paper was chosen to find a correlation between tire properties and aerodynamic drag, as a starting point for me to investigate the changes brought about by active aero on the tire properties.
Papers 1 and 2 gave me a basic grasp on the wide umbrella, that is Active Aerodynamic systems and how one would go about analyzing their effects on a land vehicle in a reliable manner. I was able to learn about what factors would affect these analyses- from vehicle geometry, ground clearance, tire housings, angle of attack, vehicle velocity in a straight line and around a corner. While I will not be able to replicate similar results in my final project, I will be going in with a better idea on what parameters I can account for and what relations I can use to obtain the results I am after.
While Paper 3 did not have a direct relation to the research project itself, it helped me establish a link between vehicle aerodynamics and tire parameters like lateral deformation, normal loads and influence on flow separation. Borrowing from the methods detailed in this paper or using the methods themselves, I am confident I can construct a link that shows the relationship between changing aerodynamic forces around a car and tire properties. This data can be useful in planning and fabrication of active aero components. On the other end, the tire impact data can also help in the construction of newer tire compounds that would be better suited to handle the extreme forces generated by active aero.

3 Technical Analysis

3.1 Introduction

In this section, I am going to talk about how I apply what I learned from the papers into a real world test case and what results I hope to extrapolate from this test case. Due to resource and time constraints, the simulations and calculations are all performed on MATLAB, relying exclusively on technical data available online and some guided, logically sound assumptions.

3.2 Methodology

For the simulations, I studied a car which was manufactured with active aerodynamic control: The 2014 McLaren P1. The P1's rear wing tucks into the rear deck at high speeds, and the adjustable underbody flaps help stabilize the car during handling and braking. This car would provide a more quantifiable means of measuring the effects of active aero than if I were to take a car that was later modified. It would also give me a reliable source for the data I will be using. Here, I mainly looked at differences in the drag coefficients between the two settings on the car; one with active aero enabled and one in a more standard "stock" mode. The dimensions of the recommended tires were also reviewed closely.

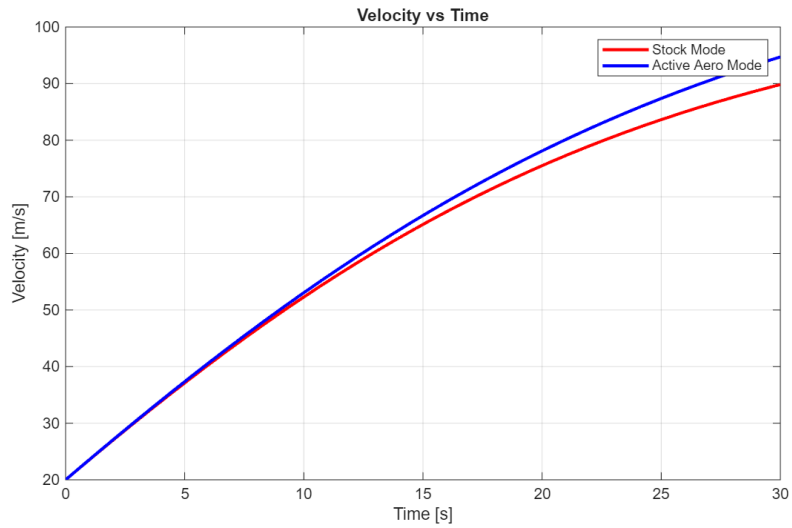
The active aero used in this vehicle was limited to a movable rear wing. This played an important role in the selection of the car, since even by the technical papers reviewed, rear wings generally provided the most "gains" in drag coefficient, and would therefore be the most ideal to analyze as a single controlled component.

With the data available to me and some educated assumptions, I ran MATLAB simulations, incorporating formulae and concepts learned in class along with those gleaned from the papers, to come to certain conclusions which are enumerated further below.

3.3 Results and Discussion

Rather than a study on distance, I have opted to study the time dependent effects of using active aero. This was so I could gauge the differences on the same timescale rather than the "difference" in distances covered between the two modes.

Straight Line Properties



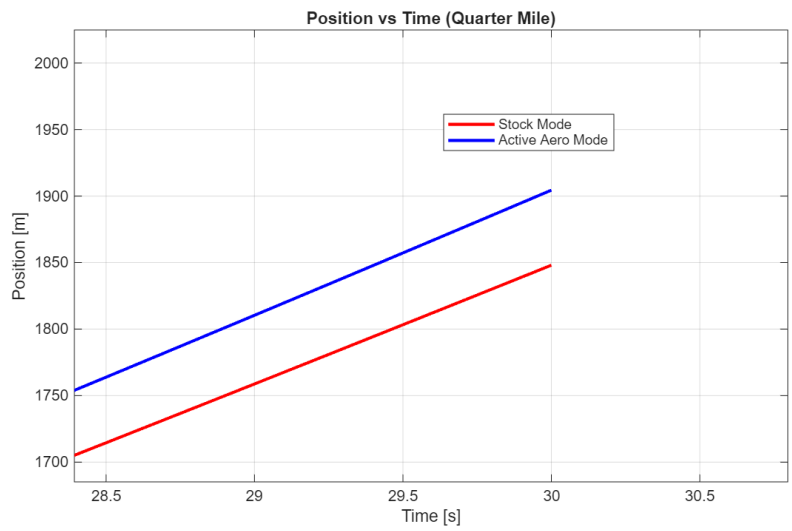


Figure 1:

The two graphs above show the benefits to the car's acceleration capabilities and velocity in figures 3.3 and 1 respectively. Fig 3.3 shows the difference in top speed at the end of the same 30 second period. It is worth noting that this first case is only assuming a straight line run for 30 seconds with a rolling start. By the end of 30 seconds, the active aero mode proved to be about 5 m/s, or 18 km/h faster than its stock counterpart.

Fig 1 further supports this, with the final distances covered at the end of 30 seconds differing by about 50 m. This demonstrates the benefit of active aero to straight line performance. The vehicle sheds drag, decreasing overall aerodynamic load, allowing the engine to dedicate more power to the moving the weight of the car itself rather than struggling against the wind as well.

Cornering and Tire Load Study

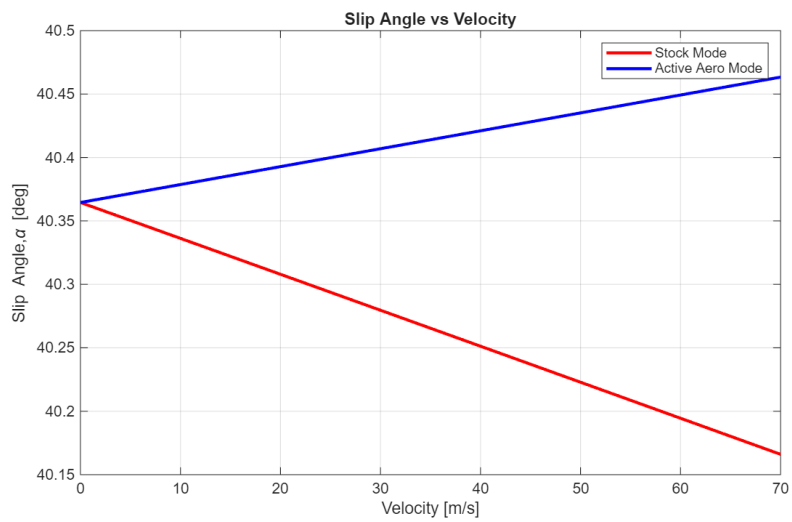


Figure 2:

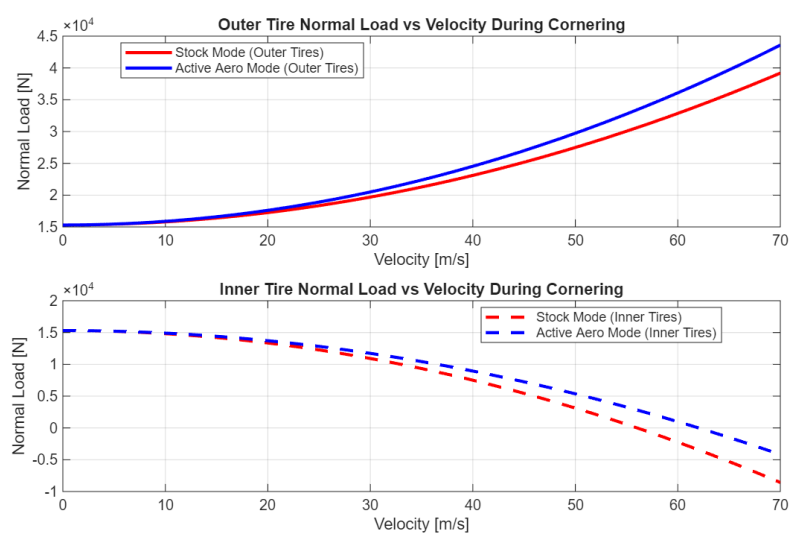


Figure 3:

The graphs above tell us a bit about the cornering properties of the car along with the effect of the excess downforce generated from the active aero systems. Fig 2 shows the slip angles of the same tires before and after the application of active aero. In stock mode, we see that at higher velocities, the slip angle decreases, indicating that it has less grip at higher velocities. Counter to this, the tires when active aero is enabled show an increase in slip angles at the same increase in velocity. This is attributed largely to the downforce generated, which in turn boosts the cornering grip.
Fig 3 reinforces this as we can see during cornering, both tires experience a higher normal load due to the downforce.

Fuel Saving

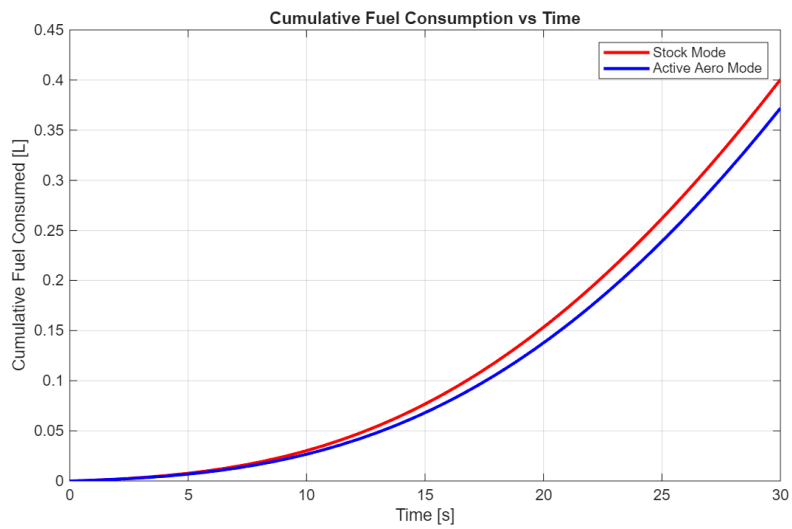


Figure 4:

Fig 4 shows that over a 30 second analysis period, the fuel consumed was little under 5mL less when the active aero kit was deployed. This being the case over a relatively smaller distance (here, a quarter mile), we have determined that over a single lap (assumed to be 5000m) that we would save 56mL of fuel and about 1.6L over a race distance of 30 laps. This is not considering the potential weight savings of loading the car with 1.6L less fuel than if it did not have active aero systems.

3.4 Conclusion

This study was mainly driven by the desire to find potential improvements in 3 areas of automobile engineering:

1. fuel economy
2. track performance
3. development and construction of tire compounds

In regards to fuel efficiency, I explored the effects of reduced aerodynamic load on an engine's fuel consumption which naturally follow the implementation of active aerodynamic systems. The on-track gains were, I believe, the most explored in this study, as they had the clearest metrics to gauge by. The reduction of drag along a straight line vs the addition of downforce along a turn were significant enough to shave between a few tenths to a second along each section of a track, summing to some significant gains over a full lap's distance.

Lastly, I investigated the effects of the aerodynamic loads on tires, since tire behavior under changing loads must be accounted for in their design and construction. The magnitude of changes in running conditions is so vast with the implementation of active aerodynamics, that in some cases, the compound used is not able to withstand the loads. Admittedly, there was not much literature on the direct relation between aerodynamic loads and changes in tire properties, so I have taken some liberties with my calculations in the form of some assumptions regarding the forces generated.
Overall, I feel this report provides a comprehensive look into the benefits of active aero systems and their influence on other fields of automobile engineering. From motorsport and tire research to optimization of daily commuter vehicles, with more possibilities unexplored in this report. Given the opportunity and resources, I would conduct a much more detailed analysis on this topic, with CFD simulations and other numerical models.

3.5 Bibliography

Car and running data:
McLaren P1 (2014)
P1 Brakes
Running data (Stock)
Tires and Tire data
Papers linked in Literature review